MTU-Former: A Multitask Unified Transformer Framework for Global Methane Monitoring With Million-Scale Multispectral Imagery

Wangyang Cao, Xiaoying Lu, Tianyi Liu, Lijie Zhang, Guangyun Zhang, Pengfei Li, and Guanghui Zhang

***Abstract*—Accurate monitoring of methane emissions is critical for addressing global climate change. While Sentinel-2 short-wave infrared (SWIR) imagery offers high spatiotemporal resolution for monitoring the dynamics of methane emissions, its complex background noise challenges automated data analysis. To address this, we propose a multitask unified transformer (MTU-Former) framework designed to simultaneously detect, segment, and quantify methane plumes by leveraging multitask synergy to enhance quantification performance. Central to our approach is the novel development of a mask-guided cross-task feature fusion (MG-CTFF) module. By fusing the plume probability map and the retrieved methane column enhancement map through a contribution map and dual physics-aware attention biases, the module guides regression attention toward core plume regions while suppressing background interference. To further resolve task discrepancies, the framework integrates a fast adaptive multitask optimization (FAMO) strategy to balance gradient updates. To support this data-driven approach, we present a large-scale dataset comprising 3 million images, constructed by combining Sentinel-2 data from 12 global regions with weather research and forecasting (WRF)-large eddy simulation (LES)-simulated plumes. Comprehensive experimental results show that MTU-Former achieves a coefficient of determination (R^2) of 0.9771, a root-mean-square error (RMSE) of 1532.93 kg h^{-1}, and a mean absolute percentage error (MAPE) of 9.55%. In pixel-level recognition, our model achieves a detection accuracy of 99.18%, a Dice coefficient of 0.9151, and an intersection over union (IoU) of 0.8673. Compared with three deep learning models, including MethaNet, Swin-UNet, and Efficient-UNet, the proposed framework reduces the RMSE by 1633.35, 562.75, and 370.47 kg h^{-1}, respectively. Correspondingly, it lowers the MAPE by 10.48%, 4.45%, and 3.09%, while improving R^2 by margins of 0.0749, 0.0199, and 0.0124. In comparison with the background-corrected integrated mass enhancement (IME) method, MTU-Former reduces the RMSE from 4323.78 to 2019.79 kg h^{-1} and MAPE from 16.76% to 9.16%, while increasing R^2 from 0.6411 to 0.9217. Overall, MTU-Former serves as a robust, versatile technical approach for significantly advancing multitask monitoring of global-scale methane emissions.**

***Index Terms*—Detection, methane plume, multitask unified transformer (MTU-Former), quantification, segmentation, weather research and forecasting (WRF)-large eddy simulation (LES).**

I. Introduction

METHANE (CH_4) is a primary driver of global climate change, possessing a 20-year global warming potential (GWP) more than 80 times higher than the GWP of carbon dioxide (CO_2) [1], [2]. It is responsible for approximately one-third of the net global warming since the Industrial Revolution [3], [4]. Compared to CO_2, methane remains in the atmosphere for only a decade. This makes methane emission control essential for meeting immediate climate goals [5]. Currently, many methane emission inventories are compiled using bottom-up approaches, which are susceptible to misreporting and uncertainties [6]. Recent large-scale observational studies have revealed significant differences between official estimates and actual emissions across various key sectors, including oil and gas production [7] and solid waste disposal [8]. Therefore, the development of robust top-down methods is essential for accurately constraining global emissions and enabling more effective atmospheric management.

Satellite remote sensing, with its extensive global coverage and frequent revisits, offers a crucial technology for the systematic monitoring of methane emissions [9]. This technique primarily relies on sensors that detect the absorption features of methane molecules in the short-wave infrared (SWIR) portion of the spectrum of sunlight reflected from the Earth's surface [10]. Although hyperspectral satellites provide the high spectral resolution needed for precise methane retrievals, they are limited by their spatial coverage and revisit frequency.

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They typically offer either coarse spatial resolution, such as Sentinel-5P [11], or narrow swath widths with long revisit times, as exemplified by PRISMA [12]. Although commercial satellites with high-resolution can accurately quantify facility emissions [13], their targeted observation modes prevent continuous global monitoring.

In contrast, multispectral satellites like Sentinel-2 [14] provide a continuous stream of data with global coverage at a 20-m spatial resolution and a 2–5-day revisit period. This makes them an ideal candidate for building high-spatiotemporal-resolution dynamic monitoring systems [15]. However, despite its potential for high-frequency global detection, Sentinel-2 possesses only a few wide bands in the SWIR spectrum. This results in a weak methane absorption signal that is highly susceptible to interference from various noise sources, rendering the robust detection and quantification of methane plumes from its imagery a significant scientific challenge [16].

Among the diverse methods developed to retrieve point-source flux rates, the integrated mass enhancement (IME) model [17], [18] stands out as the leading approach for providing high-accuracy estimations of satellite-detected plumes. Although physics-based methods like IME perform well in many scenarios, their reliance on auxiliary data, such as wind speed, and their inherent model uncertainties limit both their accuracy and their potential for full automation [19]. In recent years, deep learning has demonstrated a strong capability to automatically learn complex features from vast quantities of remote sensing data [20] and has shown promising results across various methane-related tasks. In terms of detection, Kumar et al. [21] achieved outstanding performance in automated methane plume detection using hyperspectral data from AVIRIS-NG. Furthermore, Rouet-Leduc and Hulbert [22] utilized a U-Net architecture incorporating a Vision Transformer (ViT) to process Sentinel-2 data for plume detection. Their model successfully identified emission sources as small as a few hundred kg h^{-1}, representing a significant advancement, as the sensor was previously considered effective primarily for detecting large methane plumes [16]. Beyond detection, substantial progress has also been achieved in point-source quantification. Jongaramrungruang et al. [5] employed a convolutional neural network (CNN) to quantify methane plumes, eliminating the dependency on wind speed. Radman et al. [23] further validated the effectiveness of deep learning for quantifying methane emission rates from Sentinel-2 data across three distinct regions. Additionally, Chen et al. [24] explored the problem of methane plume mask segmentation. These studies demonstrate that deep learning models achieve reliable performance on these tasks, generalizing effectively from synthetic training data to real-world plumes.

Studies in image analysis have explored multiscale feature fusion, hierarchical representations, attention mechanisms, as well as gating mechanisms for characterizing targets with weak contrast, unclear boundaries, or variable morphology [25], [26], [27], [28]. The benefits of multitask learning have been validated across a wide range of domains [29], [30]. Concurrently, Transformer architectures have demonstrated remarkable performance in remote sensing and methane retrieval tasks [22], [31]. Building on these advances and attention's strength in capturing long-range dependencies, this study explores incorporating Transformer-based modules into U-Net architectures to evaluate their effectiveness in methane multitask monitoring. We constructed a large-scale synthetic methane plume database by embedding high-fidelity plumes generated via large eddy simulation (LES) [32] into global Sentinel-2 imagery. Building upon this dataset, we propose MTU-Former, a multitask unified Transformer (MTU-Former) framework designed to leverage attention mechanisms for joint methane plume detection, pixel-level segmentation, and emission rate quantification. We systematically evaluate the framework's performance and generalization capabilities, aiming to facilitate high-frequency, top-down monitoring of methane emissions on a global scale. The main contributions of this article are summarized as follows.

1) MTU-Former is proposed as a unified multitask deep learning framework to simultaneously achieve methane plume detection, segmentation, and quantification in a single end-to-end pipeline. By unifying the disjointed multistage workflow, our model effectively mitigates error propagation inherent in cascade systems and enables global optimization. The proposed integrated approach demonstrates robustness against complex backgrounds, achieving a root-mean-square error (RMSE) of 1532.93 kg h^{-1}, R^2 of 0.9771, and mean absolute percentage error (MAPE) of 9.55% for quantification tasks, while maintaining detection accuracy of 99.18%, Dice coefficient of 0.9151, and intersection over union (IoU) of 0.8673.
2) A novel mask-guided cross-task feature fusion (MG-CTFF) module is proposed to bridge the semantic gap between segmentation and quantification. Through cross-attention guided by a methane contribution map that fuses plume probability with retrieved column enhancement, together with two learnable physics-aware attention biases grounded in plume morphology and methane mass distribution, this module enables the regression branch to focus on physically meaningful plume signals while suppressing background noise. Ablation studies demonstrate that the full MG-CTFF module reduces the RMSE by 14.33%, confirming its effectiveness for emission quantification in complex scenarios.
3) A comprehensive dataset comprising 3 million samples is constructed based on Sentinel-2 imagery from 12 distinct global regions. To the best of our knowledge, this is currently the largest global synthetic dataset for multitask methane analysis. It integrates high-fidelity LES simulations with realistic satellite noise, balancing background scenes and plume injections to ensure robust model training.

The rest of this article is organized as follows. Section II details the construction of data sources. Section III proposes our MTU-Former framework. Section IV provides the experimental results and discussion. Section V concludes this article and discusses future research directions.

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TABLE I
CONFIGURATION PARAMETERS FOR WRF-LES

Parameter	Description	Namelist Value
Surface Layer Physics	Revised Monin-Obukhov	sf_sfclay_physics = 1
Surface Flux Option	Specified Heat Flux	isfflx = 2
Diffusion Option	Full Diffusion (Physical Space)	diff_opt = 2
Eddy Coefficient	1.5-Order TKE Closure (3D)	km_opt = 2
Tracer Initialization	Passive Tracer Enabled	tracer_opt = 2
Boundary Conditions	Periodic (Lateral)	periodic_x = true periodic_y = true
Initialization Case	Idealized Simulation	ideal_case = 9

II. DATASET GENERATION

A. Methane Plume Simulation

This study employed LES to generate large-scale methane plumes with high physical fidelity. The simulations were performed using version 4.0 of the Weather Research and Forecasting (WRF) model [33]. This technique can generate the time-evolving 3-D distribution of methane within the boundary layer and allows for the specification of various environmental parameters, such as wind speed and surface heat flux. Compared to simplified models, such as the Gaussian plume model [34], WRF-LES directly resolves large turbulent motions in 3-D down to the inertial scales. This capability allows for the detailed simulation of concentration fluctuations and the formation of concentration filaments generated by vortical structures. Consequently, WRF-LES accurately reproduces the mean plume trajectories and concentration fields of passive scalar dispersion in the atmospheric boundary layer. The detailed implementation and setup of LES for simulating methane plumes have been discussed in several studies [23], [35], [36], [37].

We configured the simulation geometry to be compatible with the specifications of Sentinel-2 satellite imagery. The domain consisted of a fine-resolution inner (nested) grid and a coarser-resolution outer (driving) grid. Specifically, the inner grid had a spatial resolution of 20 m over a 2560 $\times$ 2560 m extent, while the outer grid had a resolution of 80 m over a 3200 $\times$ 3200 m extent. A one-way nesting strategy was employed to balance computational efficiency with simulation accuracy. Periodic lateral boundary conditions were applied to the outer domain to simulate the large-scale background atmospheric flow. The model output consisted of a time series of 3-D methane concentration fields. To emulate remote sensing observations, these 3-D fields were vertically integrated to produce 2-D methane column enhancement maps. Table I details the specific configuration parameters used for the WRF-LES model in this study.

Building upon the core WRF parameter configurations listed in Table I, this study focused on the critical impact of initial wind speed on methane plume morphology. Wind speed is the primary meteorological factor controlling the plume's dispersion extent, transport path, and dilution [23], [36], [38]. To systematically evaluate the effect of wind speed variations, we established 10 independent sets of WRF-LES simulations. Each set was initialized with a uniform geostrophic wind in the x-direction, with speeds ranging from 1 to 10 m s^{-1} in 1-m s^{-1} increments. Although other parameters, such as surface sensible and latent heat fluxes, also influence atmospheric stability, previous studies [15], [17], [39] have indicated that for plume morphologies represented by 2-D column enhancement maps, diverse plume shapes can be simulated using constant heat fluxes. Therefore, the latent heat flux was set to 40 W m^{-2} and the sensible heat flux to 400 W m^{-2} in these simulations, allowing us to isolate the impact of wind speed on plume structure.

In the central region of the high-resolution inner grid (20-m resolution) for each simulation, a 2 $\times$ 2 grid cell area (40 $\times$ 40 m) was selected as the methane point-source release location. A constant methane emission was introduced at this location using the WRF model's passive tracer transport capability. Each simulation was run for a total of 3 h. The first hour was designated as a spin-up period for model stabilization, ensuring that turbulence was fully developed and reached a steady state. During the subsequent 2-h simulation period, an instantaneous 3-D methane concentration field snapshot was recorded every 10 s. Upon completion, the saved 3-D concentration field data from each time step were vertically integrated. This process generated 7000 high-fidelity 2-D methane column enhancement maps. With the help of WRF-LES, the simulated plumes capture the influence of turbulent dispersion, so the ten wind speeds yield distinct morphology in direction, length, and internal filament structure. With later emission scaling, rotation, and translation, the plumes are not restricted to a single shape.

Given that the simulation sensitivity is independent of linear scaling [39], the simulation outputs were scaled to an emission range of 5000–30 000 kg h^{-1}. This interval is specifically tailored to Sentinel-2's performance characteristics, bridging the detection gap between ultralarge emitters identified by Sentinel-5P and smaller sources captured by airborne surveys [18], [40], [41]. To further ensure consistency with practical observational constraints, we applied a detection threshold based on the multiband multipass (MBMP) precision of $\approx$0.13 mol m^{-2} ($\sim$0.365 ppm) [18], masking pixels below this limit as background when identifying valid plume signals. Fig. 1 presents examples of the resulting simulated plumes across varying wind speeds and emission rates.

B. Sentinel-2 Real Noise

Background clutter is a critical factor influencing methane plume monitoring tasks [15], [23]. We constructed our dataset using actual plume-free Sentinel-2 multispectral imagery to ensure the synthetic data faithfully represents real-world observation conditions. This approach is essential because standard Gaussian noise models fail to replicate the spectral complexities introduced by vertical urban structures. Neglecting these environmental factors reduces the realism of the synthetic data and can hinder the performance of subsequent processing tasks [42].

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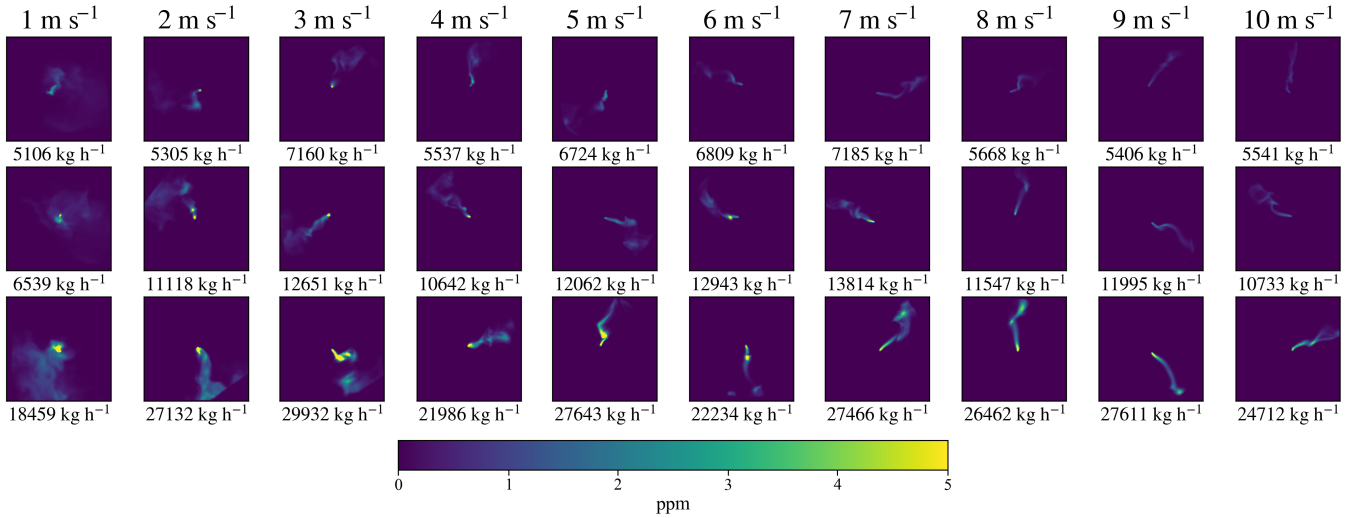


Fig. 1. WRF visualizations of methane column enhancement plumes with the top, middle, and bottom rows corresponding to low, medium, and high emission rates, respectively.

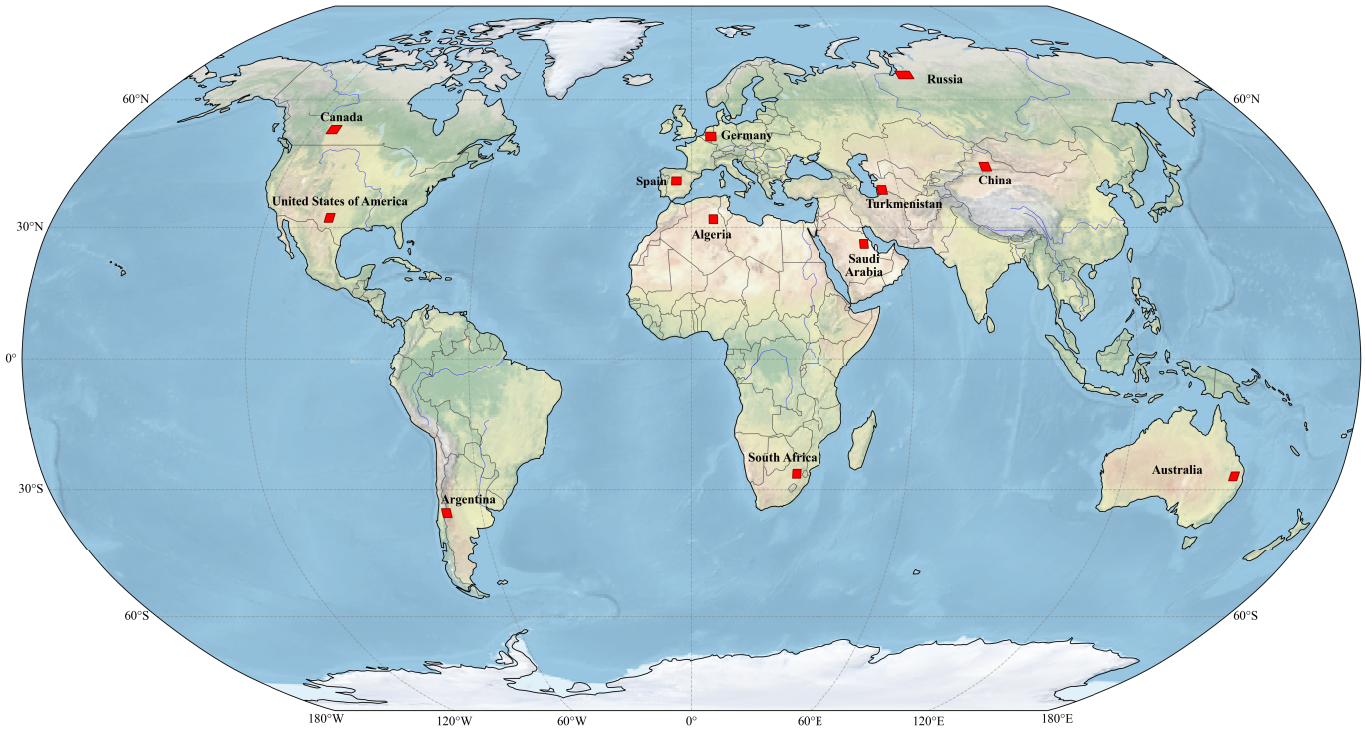


Fig. 2. Global distribution of the 12 study areas located in Algeria, Spain, Turkmenistan, the United States, Russia, China, Canada, South Africa, Germany, Saudi Arabia, Australia, and Argentina.

To construct the background component, we processed Sentinel-2 imagery spanning 2022–2023 across 12 global regions exhibiting high-frequency methane plumes, including Algeria, Spain, Turkmenistan, the United States, Russia, China, Canada, South Africa, Germany, Saudi Arabia, Australia, and Argentina. To reduce cloud contamination before the MBMP retrieval, we retained only Sentinel-2 level 1 C (L1C) scenes with cloud coverage below 1%. Methane plumes in these areas have been documented in prior studies [22], [43]. These regions exhibit diversity and representativeness in terms of topography, natural environment, and geographic coordinates. They serve as regional exemplars, ensuring the global generalizability of the dataset. Fig. 2 shows the specific regions we selected.

In this study, we employed the MBMP method [18], which reduces retrieval artifacts compared with single-pass methane retrieval. This method utilizes Sentinel-2's SWIR Band 12 (~2190 nm) and Band 11 (~1610 nm) as the primary retrieval channels [15]. Band 12 provides the main methane-sensitive signal due to stronger methane absorption

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in the SWIR region, while Band 11, with weaker methane absorption, serves as a reference channel for reducing background effects. Together with the diverse real Sentinel-2 backgrounds described above, this channel pair helps preserve physically meaningful methane-sensitive information while accounting for regional background variations. The core of the MBMP method lies in combining multiband information with multitemporal observations to correct systematic errors in single-pass retrievals by removing a similar plume-free reference. The retrieval process first requires calculating the single-pass multiband reflectance ratio change, ΔR_{MBSP}, formulated as [18]

$$\Delta R_{MBSP} = \frac{cR_{12} - R_{11}}{R_{11}} \tag{1}$$

where R_{12} and R_{11} are the apparent reflectances of Band 12 and Band 11, respectively, and c is a scaling factor used to correct for scene brightness differences. The final MBMP methane column enhancement $\Delta\Omega_{MBMP}$ is obtained through differencing to eliminate background interference

$$\Delta\Omega_{MBMP} = \Delta\Omega_{MBSP} - \Delta\Omega'_{MBSP} \tag{2}$$

in which $\Delta\Omega_{MBSP}$ and $\Delta\Omega'_{MBSP}$ represent the retrieved methane column enhancements for the target scene and the plume-free reference scene, respectively.

To accurately and efficiently retrieve methane concentrations from the observed signals, we employed a precomputed look-up table (LUT) method [15]. Building on prior research, we used the radiative transfer model (Libradtran) [44] to establish the physical relationship among methane transmission, methane column concentration enhancement ΔX_{CH_4} (ppm), and air mass factor (AMF) under various observation geometries. This relationship, based on the Beer–Lambert law, can be approximated as [39]

$$T_{plume}(\lambda) \approx e^{-AMF\cdot\sigma_{CH_4}\cdot\Delta X_{CH_4}} \tag{3}$$

where σ_{CH_4} is the methane absorption cross section. We further convolved the simulated results with the spectral response functions (SRFs) of Sentinel-2 Band 11 and Band 12. This process established the quantitative relationship between the observed radiance change and the actual methane concentration enhancement.

C. Dataset Construction

After obtaining Sentinel-2 L1C data for the selected study regions, we employed the MBMP retrieval method to generate a background dataset. For each tile, we paired images with a temporal interval of no more than 60 days. Each image pair was processed using a 128 × 128 pixel sliding window with no overlap. Within each window, the original low-resolution solar zenith angle (SZA) and viewing zenith angle (VZA) were bilinearly interpolated to a 20-m resolution before the mean angular values were calculated. This resampling was applied only to the observation-geometry layers to match the 20-m spatial grid of Sentinel-2 Bands 11 and 12 used in the MBMP retrieval, with bilinear interpolation chosen for the smoothly varying SZA and VZA fields. Based on these mean angles and the precomputed LUT, we performed the MBMP retrieval on each window to obtain the corresponding methane column concentration enhancement map.

The retrieved subwindows were then subjected to a statistical filtering process to exclude potential, unidentified real methane signals. According to the outlier rejection scheme of Radman et al. [23], we excluded the background subwindows with mean concentrations exceeding the global mean plus 1.65 standard deviations. This measure was intended to ensure that the background dataset used for subsequent synthesis was as free as possible of anomalous methane signals. The resulting satellite background images are shown in Fig. 3.

To enhance dataset diversity during synthesis, we applied random rotation (0°–359°) and translation (±5 pixels) to the scaled WRF-LES simulated plumes and masks before superimposing them onto real background noise images. To ensure realistic detectability, we implemented a filtering condition requiring the plume's core signal to exceed the background noise standard deviation. This constraint guarantees a sufficient signal-to-noise ratio (SNR) for robust identification [45]. Fig. 4 shows examples of the resulting synthetic data, illustrating various regions, wind speeds, and distinct SNRs. For training, the regression labels are linearly scaled to [0, 1] over the range 0–30 000 kg h^{-1} and mapped back to physical units before evaluation.

III. Methodologies

A. Proposed MTU-Former Framework

To address the challenges of weak methane plume signals, complex background interference, and variable morphology in Sentinel-2 satellite imagery, this study proposes the multitask unified deep learning framework MTU-Former. This model mitigates stepwise error propagation and eliminates wind speed dependency. It refines quantification by leveraging a mask-guided feature fusion strategy to impose plume morphological boundaries as physical constraints.

The input to MTU-Former is a 128 × 128 single-channel methane enhancement map, derived from Sentinel-2 Bands 11 and 12 through MBMP retrieval followed by LUT-based conversion. This preprocessing step embeds the spectral information into the methane enhancement map, which is used as the direct network input instead of raw multispectral bands. The model is based on the U-shaped CSWinUNet [46] architecture. Using a patch size of 4 with patch normalization, the input is embedded into 32 × 32 tokens without absolute position embedding. The CSWin Transformer [47] encoder contains four hierarchical stages with depths of [2, 2, 6, 2], embedding dimensions of [96, 192, 384, 768], attention heads of [3, 6, 12, 24], window sizes of [4, 4, 4, 4], and an multilayer perceptron (MLP) ratio of 4. The decoder is arranged in the reverse order with depths of [2, 6, 2, 2], using patch-expanding operations and skip connections to recover high-resolution spatial details. The segmentation and detection heads both output two classes, while the quantification branch uses the MG-CTFF module with a 256-D fusion space and 8 attention heads before the regression head. As shown in Fig. 5, unlike standard models [48], spatial dependencies are captured using a cross-shaped

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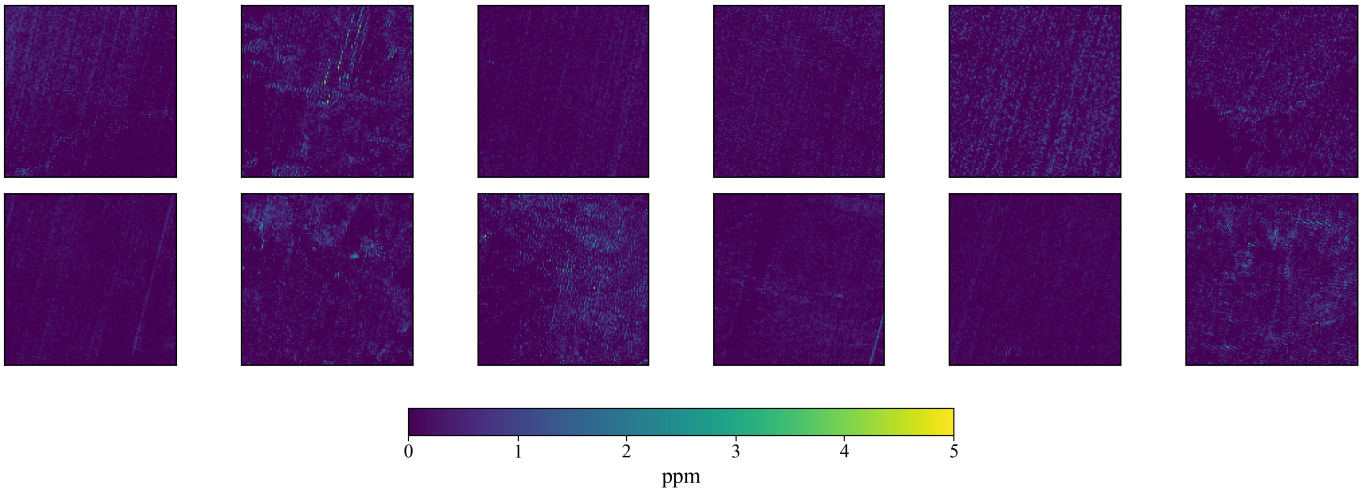


Fig. 3. Representative samples from the constructed global satellite background dataset. The panels display methane column concentration enhancements retrieved from Sentinel-2 data, selected from 12 representative regions worldwide to demonstrate background noise without active plumes.

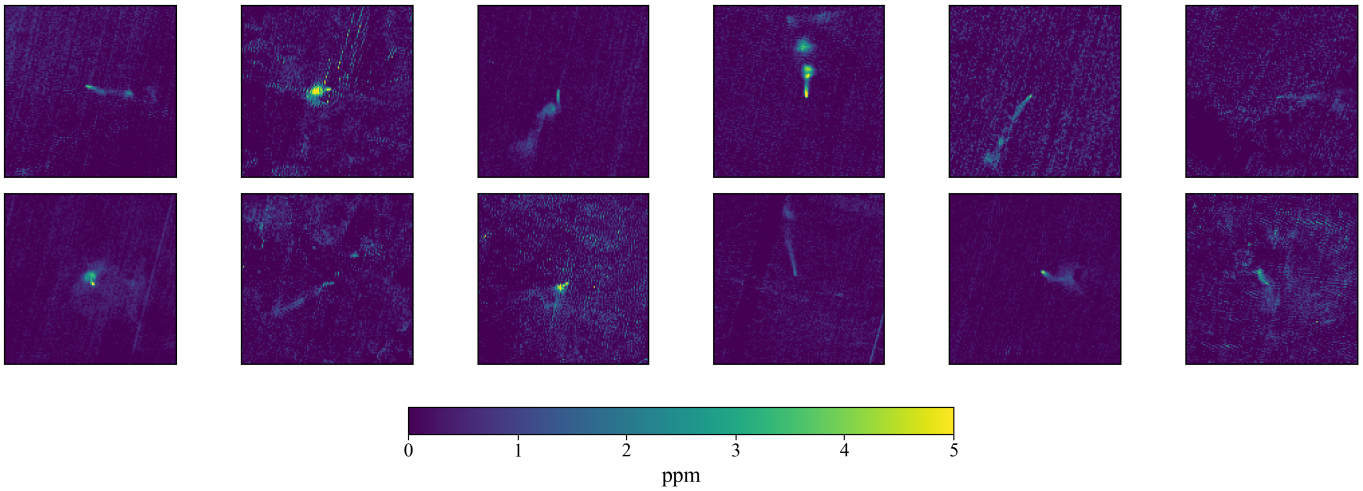


Fig. 4. Examples of the synthetic dataset combining the background noise defined in Fig. 3 with WRF-LES simulated methane plumes. Data augmentation is applied to introduce variability in plume orientation and intensity.

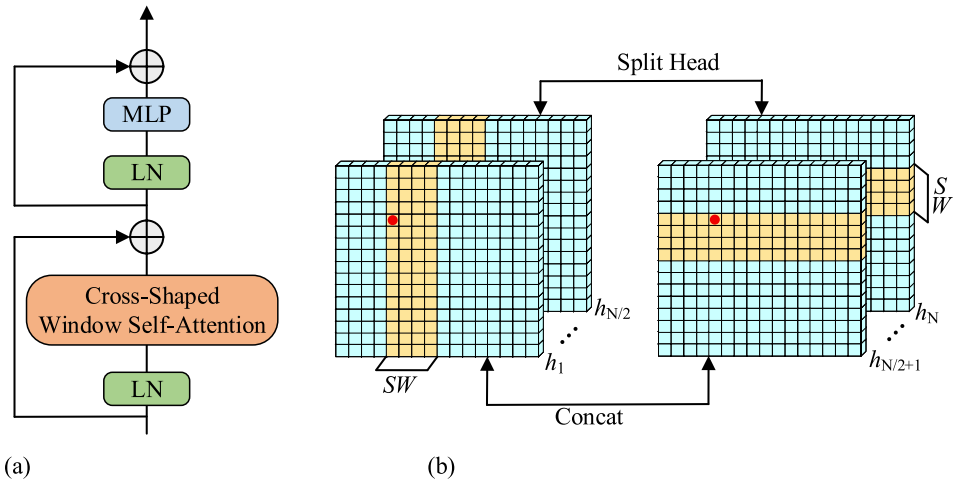


Fig. 5. Components of the CSWin Transformer. (a) CSWin Transformer block. (b) Diagram of cross-shaped window self-attention with split-head processing.

window self-attention mechanism. By computing attention along horizontal and vertical stripes, this approach effectively expands the receptive field to model long-range interactions. Fig. 6 illustrates the overall procedure.

To clarify how the different tasks are structurally connected and integrated, all tasks in our framework first share the same CSWin Transformer-based feature encoder to extract multiscale generalized features. Building upon this shared

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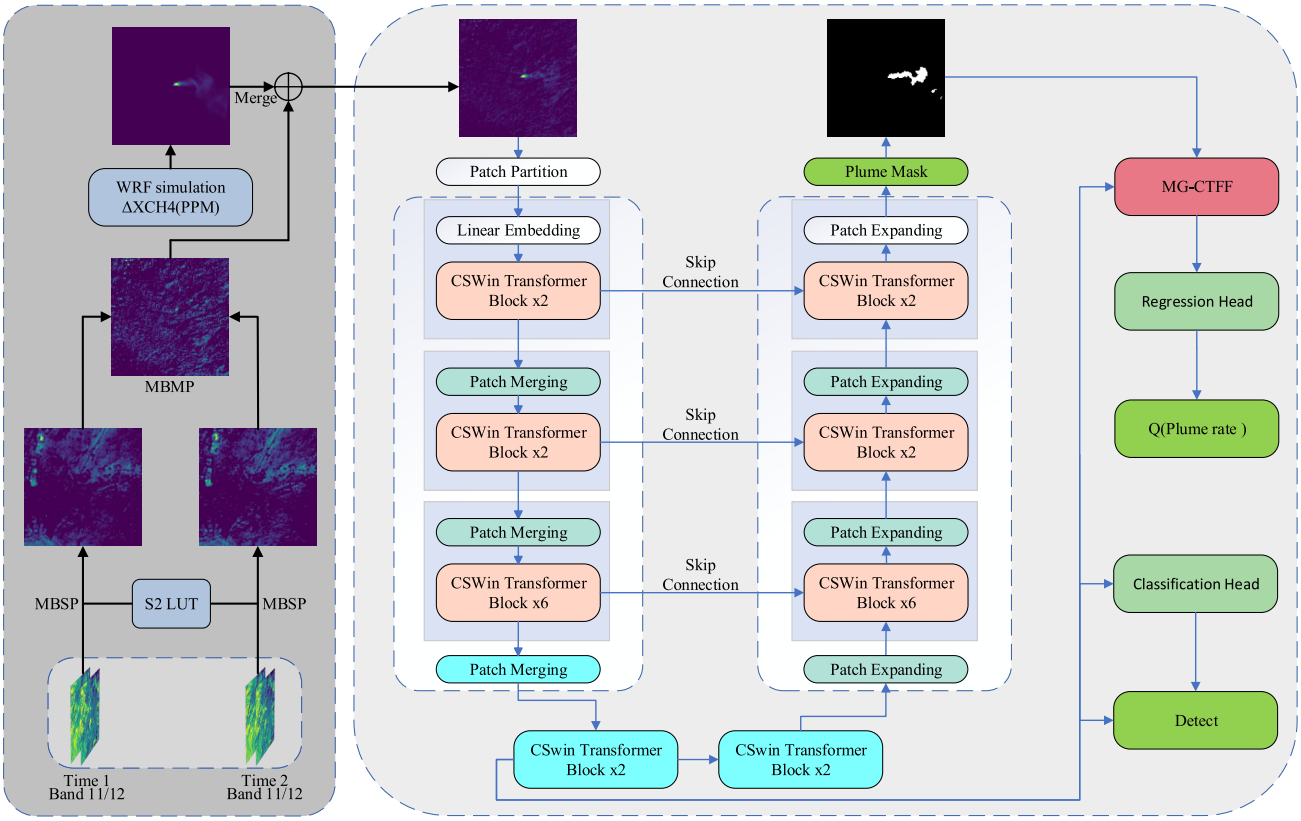


Fig. 6. Schematic of the proposed MTU-Former based on the CSWinUNet architecture. The framework performs simultaneous detection, segmentation, and quantification regression tasks, incorporating the MG-CTFF fusion module before the regression head.

bottleneck, the three tasks employ distinct integration strategies. Specifically, the detection task directly utilizes the global bottleneck features for binary classification. The segmentation task employs the mirrored decoder paths to upsample the bottleneck features, recovering high-resolution spatial details to output a pixel-level mask. In contrast, the integration of the quantification task is the most profound. Rather than operating in isolation, it utilizes our proposed MG-CTFF module. Inspired by the cross-attention mechanism in [49], our module uniquely performs multiscale cross-attention fusion between the segmentation probability map output by the segmentation branch and the bottleneck features. This design allows the quantification task to directly leverage the physical boundary priors from the segmentation task, effectively suppressing background noise and ultimately achieving deep intertask coupling.

B. Feature Encoding and Adaptive Optimization Strategy

To capture methane plume morphology and ensure balanced task convergence, we employ a Transformer-based backbone combined with dynamic multitask optimization. Distinct from discrete surface objects, methane plumes in Sentinel-2 imagery typically appear as diffuse, low-contrast features that extend downwind. Traditional CNNs are restricted by fixed local receptive fields, while standard Swin Transformers [48] rely on square window attention, which often constrains their efficiency in capturing long-range directional dependencies.

To address this, we adopt the CSWin Transformer [47] as the feature encoder. The core advantage of this architecture lies in its cross-shaped window self-attention mechanism, which decouples self-attention into parallel horizontal and vertical stripe branches. This design naturally aligns with the physical diffusion characteristics of methane plumes, enabling the model to establish long-range spatial dependencies spanning kilometers within a single block. By processing features along orthogonal directions, the encoder preserves the continuity of weak plumes while effectively filtering out background noise. This hierarchical structure generates multiscale feature representations rich in semantic context, providing a robust foundation for the subsequent detection, segmentation, and quantification branches.

To simultaneously optimize all tasks, we formulate the joint training as a multiobjective optimization problem, where the total loss $\mathcal{L}_{\text{total}}$ is defined as a weighted sum of the task-specific losses

$$\mathcal{L}_{\text{total}} = w_{\text{det}}\mathcal{L}_{\text{CE}}^{\text{det}} + w_{\text{seg}}\left(\mathcal{L}_{\text{CE}}^{\text{seg}} + \mathcal{L}_{\text{Dice}}^{\text{seg}}\right) + w_{\text{reg}}\mathcal{L}_{\text{MSE}}^{\text{reg}} \quad (4)$$

where $\mathcal{L}_{\text{CE}}^{\text{det}}$ denotes the cross-entropy loss for detection and $\mathcal{L}_{\text{mse}}^{\text{reg}}$ represents the mean-squared error loss for regression. For the segmentation task, we use a hybrid objective combining the pixel-wise cross-entropy loss $\mathcal{L}_{\text{CE}}^{\text{seg}}$ and the shape-aware Dice

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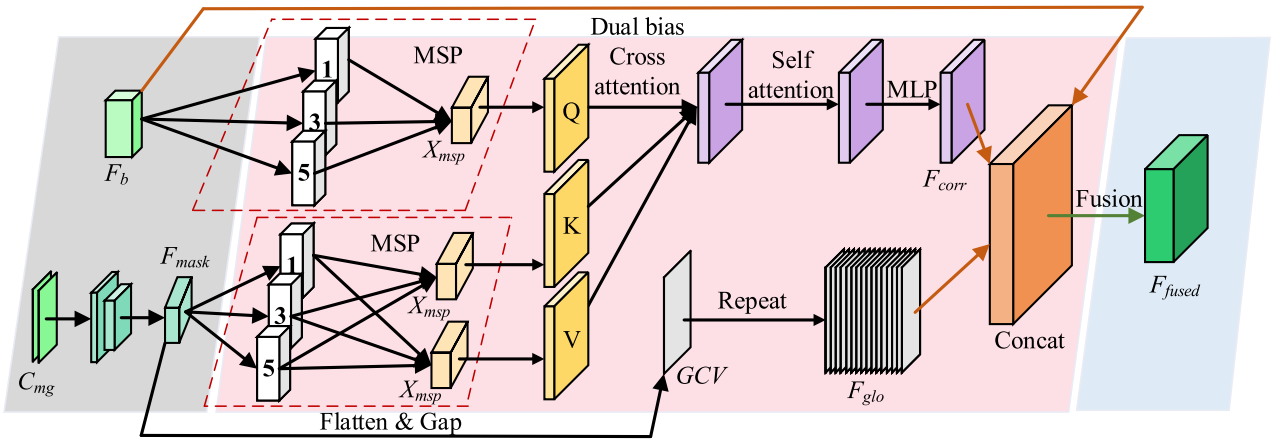


Fig. 7. Structure of the MG-CTFF module.

loss $\mathcal{L}_{\text{Dice}}^{\text{seg}}$. Due to the significant heterogeneity in gradient magnitudes and convergence speeds among the quantification (mse loss), segmentation (CE + Dice loss), and detection (CE loss) tasks, traditional linear weighting often leads to certain tasks dominating the training process. Therefore, the balancing coefficients w_{det}, w_{seg}, and w_{reg} are not fixed hyperparameters. Instead, we utilize the fast adaptive multitask optimization (FAMO) strategy [50]. In each forward pass, after computing the stacked task losses, FAMO dynamically calculates and updates the individual weights before backpropagation based on the relative descent rates of each task's loss. This dynamic adjustment balances gradient updates, prevents the model from falling into local optima, and ensures robust joint convergence across the multitask framework.

C. MG-CTFF for Plume Quantification

While the CSWin Transformer encoder extracts essential plume features, its direct integration with parallel decoders introduces background noise that impairs source strength estimation. To alleviate this issue, we propose the MG-CTFF module to impose a physical consistency constraint between the segmentation and regression tasks.

The underlying logic of this approach follows the workflow of the expert annotation process, where the precise physical boundaries of the plume are outlined before the emission rate is computed. Accordingly, we utilize the mask probability map produced by the segmentation branch not merely as a final output but as a form of prior knowledge embodying morphological topological information. Beyond this, the module further incorporates the retrieved methane column enhancement to construct a methane contribution map that reflects the per-pixel mass distribution within the plume. This structural and physical information is deeply integrated into the regression branch through a modified cross-attention feature fusion module [49] equipped with dual physics-aware attention biases. This architecture directs the network to focus on regions with valid concentration patterns and significant methane loading, helping to reduce noise caused by surface reflectance heterogeneity.

As illustrated in Fig. 7, the fusion mechanism acts upon two heterogeneous input streams comprising the bottleneck features $F_b \in \mathbb{R}^{L\times C}$ derived from the deep encoder layers and the predicted mask logits $M_{seg} \in \mathbb{R}^{H\times W\times N_{class}}$ obtained from the segmentation head. In this formulation, L represents the length of the flattened spatial token sequence, C is the channel dimensionality, $H \times W$ denotes the spatial resolution, and N_{class} indicates the number of semantic classes. In this dual-stream architecture, F_b extracts key features for plume quantification, and M_{seg} identifies the region extent of plumes. We maintain full gradient connectivity between the regression module and the segmentation head. In this end-to-end approach, the regression objective acts as an auxiliary constraint, guiding the segmentation branch to learn features that are both semantically discriminative for boundaries and physically informative for quantification. To further incorporate the per-pixel methane mass distribution into the regression branch, we construct a methane contribution map $G_{\text{contribution}} \in \mathbb{R}^{H\times W\times 1}$ through element-wise multiplication between the predicted plume probability and the retrieved methane column enhancement

$$G_{\text{contribution}} = P_{\text{plume}} \odot \Delta X_{\text{CH}_4} \tag{5}$$

where P_{plume} is the per-pixel plume class probability obtained by applying the Softmax function to the segmentation logits M_{seg} and ΔX_{CH_4} denotes the MBMP-retrieved methane column enhancement. This design reflects the intuition that each pixel's contribution to the total plume mass is jointly determined by its probability of belonging to the plume and the magnitude of its methane enhancement. Pixels with both high plume confidence and large methane enhancement naturally receive stronger emphasis, while background pixels or those with low enhancement are suppressed. We then design a lightweight mask feature processor. Utilizing a two-layer strided convolution, we project the concatenated mask-contribution input $C_{mg} = [M_{seg}, G_{\text{contribution}}]$ into a compressed semantic space, thereby generating the mask context feature F_{mask}. This operation aligns spatial resolution and encodes both pixel-level morphological confidence and

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methane mass distribution into a unified context for the regression task.

Given that methane plumes exhibit multiscale characteristics ranging from localized point-sources to extensive long-distance dispersion, single-scale feature interaction is insufficient to fully capture complex morphological correlations. Inspired by the CAFF module [49], we design a feature-level multiscale projection (MSP) strategy. Unlike the original CAFF, which computes multiple attention maps independently, we employ a set of parallel convolutional kernels $k_i \in \{1, 3, 5\}$ to project both the query and context features prior to their entry into the attention mechanism. This approach captures local details and global structure across varying receptive fields. These multiscale features are concatenated along the channel dimension, flattened, and subsequently projected into a unified embedding space via a linear layer, yielding the multiscale projected feature representation $\mathbf{X}_{msp}$ which encapsulates spatial contexts from varying receptive fields

$$\mathbf{X}_{msp} = \mathcal{L}_{msp}\left(\mathcal{F}_{flat}\left(\underset{k\in\{1,3,5\}}{\text{Concat}}\left[\text{Conv}_k\left(\mathbf{X}_{in}\right)\right]\right)\right) \tag{6}$$

where $\mathbf{X}_{in}$ represents the input feature map of MSP. $\text{Conv}_k(\cdot)$ denotes a 2-D convolution with kernel size k and Concat$[\cdot]$ represents channel-wise concatenation. $\mathcal{F}_{flat}(\cdot)$ is the flattening operation that transforms spatial dimensions into a token sequence and $\mathcal{L}_{msp}(\cdot)$ is a linear layer that projects the fused features into the target embedding dimension.

Subsequently, this MSP strategy is independently applied to both the bottleneck features and the mask features to map them into a unified latent space for interaction. Specifically, the bottleneck features F_b processed by the MSP module serve as the query (Q), while the MSP-processed mask features F_{mask} act as both the key (K) and value (V). These embeddings are fed into a composite attention module. To make physical priors directly modulate the attention distribution, we replace the standard cross-attention with a physics-biased cross-attention mechanism. Before attention computation, the plume probability map and methane enhancement map are resized and flattened to the same token resolution as the key features. The plume morphology bias and methane contribution bias are defined as

$$\tilde{C}_j = \frac{\text{ReLU}\left(\Delta X_{\text{CH}_4,j}\right)}{\sum_k \text{ReLU}\left(\Delta X_{\text{CH}_4,k}\right) + \epsilon} \tag{7}$$

$$b_j^P = \log\left(P_j + \epsilon\right) \tag{8}$$

$$b_j^C = \log\left(\tilde{C}_j + \epsilon\right) \tag{9}$$

where P_j is the plume probability of the jth token, $\tilde{C}_j$ is the sum-normalized positive methane enhancement, and ϵ is a small constant for numerical stability, set to 1×10^{-3} in our implementation. The modified attention score for the hth head is then

$$S_{ij}^{(h)} = \frac{Q_i^{(h)}\left(K_j^{(h)}\right)^\top}{\sqrt{d_h}} + \lambda_h^P b_j^P + \lambda_h^C b_j^C \tag{10}$$

where d_h denotes the dimensionality of each attention head and λ_h^P and λ_h^C are learnable per-head scalars controlling the strengths of the plume morphology and methane contribution priors, respectively. They are initialized to 0.55 and 0.35, excluded from weight decay, and optimized with a learning rate ten times the base rate, allowing the two physical priors to adapt their relative influence during training. The resulting attention weights are obtained by applying Softmax over the key-token dimension. In this way, the cross-attention is encouraged to focus on plume-like regions with stronger methane loading. Physics-biased cross-attention is followed by a residual self-attention block to enhance internal context, and finally an MLP to yield the final correlation map $\mathbf{F}_{corr}$. Global average pooling (GAP) is applied to the mask features F_{mask} to extract the global context vector (GCV), which captures the entire plume mask information. The GCV is repeated along the sequence dimension to align with the bottleneck features, producing the expanded global representation $\mathbf{F}_{glo}$.

Finally, to achieve precise emission rate quantification, we propose a three-branch feature fusion strategy, where the aggregated feature comprises three components.

1) *The Original Bottleneck Feature* $\mathbf{F}_b$. It is intended to preserve the uncompressed high-dimensional concentration intensity representation and spatial semantic information, serving as the fundamental physical basis for the regression task to infer emission magnitude.
2) *The Correlation Map* $\mathbf{F}_{corr}$. Here, the bottleneck feature $\mathbf{F}_b$ is spatially adjusted based on shape details. Through this mask-guided approach, the model emphasizes signals within valid plume boundaries while attenuating background interference.
3) *The Global Mask Context* $\mathbf{F}_{glo}$. By embedding global plume confidence into feature tokens as a contextual prior, the model can better handle sparse or plume-free cases.

After extracting these components, we concatenate them along the channel dimension and employ a learnable linear projection layer to synthesize the heterogeneous information into a unified space. The final fusion process is formulated as

$$\mathbf{F}_{\text{fused}} = \mathcal{L}_{fuse}\left(\text{Concat}\left[\mathbf{F}_b, \mathbf{F}_{\text{glo}}, \mathbf{F}_{\text{corr}}\right]\right) \tag{11}$$

where Concat$[\cdot]$ denotes the concatenation operation along the channel axis. $\mathcal{L}_{fuse}(\cdot)$ projects the concatenated features into a compact embedding space. The resulting tensor $\mathbf{F}_{\text{fused}}$ encodes both local semantic details and global contextual priors, serving as the direct input to the subsequent regression head for methane quantification.

IV. Results and Discussion

A. Experimental Configuration and Parameter Settings

All experiments were implemented in PyTorch using distributed data parallel (DDP) on 6 Alibaba ZW810E parallel processing units (PPUs). The synthetic dataset contained 3 million methane column enhancement maps with a fixed input size of 128×128 and was randomly divided into training, validation, and testing sets at a ratio of 7:2:1. MTU-Former was trained for 60 epochs with a batch size of 128 per PPU, corresponding to a global batch size of 768. The base learning rate was set to 1×10^{-5} and linearly scaled by the number of PPUs according to the linear scaling rule

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[51], resulting in an effective peak learning rate of 6×10^{-5}. A 10-epoch linear warmup was followed by cosine annealing with a minimum learning rate of 1×10^{-6}. AdamW was used as the optimizer with a weight decay of 0.1. For regularization, DropPath was linearly increased from 0 to 0.2 across the encoder blocks and then mirrored in the decoder. No global dropout or backbone attention dropout was used, except for a dropout rate of 0.1 in the cross-task attention fusion module. For FAMO, the regularization coefficient was set to 0.001, the task-logit learning rate for updating task weights was set to 0.025, and gradient clipping was applied with a maximum norm of 1.0. The checkpoint with the lowest validation RMSE was selected for testing.

B. Evaluation Metrics

To comprehensively assess MTU-Former, we employed standard metrics across three domains: detection, segmentation, and quantification.

For the binary detection task, we reported accuracy to measure the global correctness of plume identification, providing a reliable foundation for subsequent processing, which is defined by

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

To evaluate pixel-level morphological consistency, we utilized the Dice similarity coefficient and IoU. Methane plumes usually cover a small part of the image; Dice and IoU are more informative than pixel-wise accuracy because they focus on plume-region overlaps and penalize both missed plume areas and false plume predictions

$$\text{Dice} = \frac{2\,|P \cap G|}{|P| + |G|}, \quad \text{IoU} = \frac{|P \cap G|}{|P \cup G|} \tag{13}$$

where P denotes the binary prediction mask, while G represents the ground truth.

For emission rate quantification, we assessed performance using the RMSE, MAPE, and coefficient of determination (R^2). The RMSE measures the absolute error in physical emission scale, the MAPE reports the relative error across different emission levels, and R^2 reflects the overall agreement between predicted and reference emission variations. These are calculated by

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{14}$$

$$\text{MAPE} = \frac{100\%}{N_p}\sum_{i=1}^{N_p}\frac{|y_i - \hat{y}_i|}{y_i} \tag{15}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{16}$$

where y_i represents the ground truth, $\hat{y}_i$ is the predicted value, and $\bar{y}$ denotes the mean of the observed data. N is the total number of samples. To ensure the MAPE is defined, N_p counts only the samples with nonzero emissions.

C. Results on Synthetic Dataset

Given that research on the multitask quantification of methane plumes using satellite data is still in its initial stage, a unified benchmark for direct comparison remains absent in the literature. To provide a comprehensive and objective assessment of the performance of MTU-Former, we established a comparative framework comprising traditional physical methods, single-task deep learning models, and purposefully enhanced strong multitask baselines. The IME method [18] was selected to represent traditional physical quantification techniques, while MethaNet [5] served as the single-task deep learning baseline. To evaluate the multitask architecture, we compared MTU-Former against four U-shaped baselines: the CNN-based EfficientNetV2-L-UNet [52], the Transformer-based Swin-UNet [53], ViT-UNet [22], and MPSUNet [24]. All multitask models adopted the same U-shaped decoder structure and were trained with the FAMO strategy, ensuring that performance differences reflect the backbone architecture and feature fusion design rather than training configuration.

Table II summarizes the performance of each model on the full synthetic dataset. MTU-Former achieved the lowest RMSE (1532.93 kg h^{-1}), MAPE (9.55%), and highest R^2 (0.9771), indicating the best quantification accuracy. MPSUNet obtained the highest Dice (0.9312), IoU (0.8882), and detection accuracy (99.54%), benefiting from its domain-specific design for methane plume segmentation. Among the general-purpose baselines, EfficientNetV2-L-UNet and Swin-UNet delivered competitive results, whereas ViT-UNet underperformed across all metrics, potentially reflecting the limited local inductive biases of the encoder. Additionally, scatter plots comparing predicted and ground-truth emission rates for all models on the 150 000 test samples are provided in the Supplementary Material.

All multitask models achieved satisfactory quantitative performance on the segmentation task. Fig. 8 further presents MTU-Former's predictions across twelve representative regions, showing that simple plume bodies and their wind-driven elongated shapes are largely recovered. But for plumes with complex structures or weak contrast, the distal tail regions and fine-scale details are still more prone to boundary deviations and fragmented predictions. The key distinction lies in quantification precision, where MTU-Former reduced the RMSE by 6.4% relative to the second-best method (MPSUNet). MPSUNet attains the highest segmentation scores but not the best quantification accuracy, indicating a clear gap between the two tasks and the necessity of bridging them at the feature level. This improvement is attributed to the MG-CTFF module and CSWin Transformer backbone, which jointly enhance cross-task feature interaction and help the model focus on valid plume regions for more accurate emission estimation.

Table III further compares the computational cost of each model. MTU-Former, MethaNet, and Swin-UNet were trained and evaluated with the native $1 \times 128 \times 128$ input. For Efficient-UNet, ViT-UNet, and MPSUNet, which are originally designed for multichannel inputs, the same single-channel map was repeated along the channel dimension to form a

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TABLE II

PERFORMANCE COMPARISON UNDER SETTING A: AUTOMATED MONITORING (TESTED ON FULL DATASET)

Method	RMSE (kg h^{-1})	MAPE (%)	R^2	Dice	IoU	Detection Accuracy (%)
MethaNet	3166.28	20.03	0.9022	–	–	–
Efficient-UNet	1903.40	12.64	0.9647	0.8909	0.8443	99.28
Swin-UNet	2095.68	14.00	0.9572	0.9148	0.8662	99.09
ViT-UNet	2937.64	17.39	0.9159	0.8243	0.7673	95.44
MPSUNet	1638.50	10.84	0.9738	**0.9312**	**0.8882**	**99.54**
MTU-Former	**1532.93**	**9.55**	**0.9771**	0.9151	0.8673	99.18

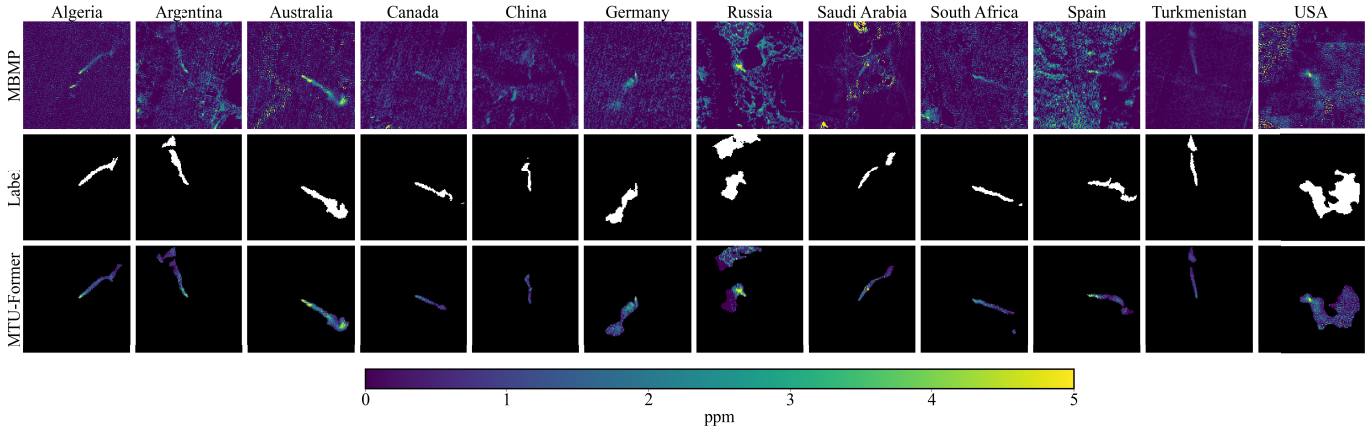


Fig. 8. Qualitative comparison between MTU-Former segmentation results and mask labels across 12 representative regions. The overall RMSE of emission-rate prediction across these cases is 1276.86 kg h^{-1}.

TABLE III

COMPUTATIONAL EFFICIENCY COMPARISON OF ALL EVALUATED MODELS

Method	Params (M)	FLOPs (G)	FPS (img s^{-1})	Inference Time (ms)	Peak Memory (MB)	Training Throughput (img s^{-1})
MethaNet	**0.183**	**0.024**	**3358.10**	**0.298**	**11.19**	**8639.73**
Efficient-UNet	122.690	4.788	54.33	18.409	504.36	467.71
Swin-UNet	56.054	3.064	94.38	10.596	239.77	712.08
ViT-UNet	57.416	13.622	193.04	5.180	264.56	367.79
MPSUNet	25.034	13.049	255.65	3.912	137.13	255.86
MTU-Former	57.946	3.126	49.21	20.326	246.14	563.73

TABLE IV

PERFORMANCE COMPARISON UNDER SETTING B: QUANTIFICATION FOCUS (TESTED ON PLUME-PRESENT SUBSET)

Method	RMSE (kg h^{-1})	MAPE (%)	R^2	Dice	IoU	Detection Accuracy (%)
IME (Ideal)	2349.78	10.68	0.8940	–	–	–
IME (Bg-Corrected)	4323.78	16.76	0.6411	–	–	–
MTU-Former	**2019.79**	**9.16**	**0.9217**	**0.8480**	**0.7503**	**98.95**

3 × 128 × 128 input. MethaNet is the most lightweight model but lacks multitask capability. Efficient-UNet has the largest parameter count, while Swin-UNet and ViT-UNet share similar parameter scales but different in FLOPs due to their distinct attention designs. MPSUNet achieves higher segmentation metrics and faster inference with fewer parameters, making it more suitable for real-time detection. In contrast, MTU-Former requires only about one-quarter of the FLOPs of MPSUNet and delivers higher quantification accuracy than all compared models, making it more suitable for precise emission estimation.

For the physical method, to ensure a rigorous comparison with the IME method, we restricted the testing scope to the plume-present subset, even though our MTU-Former

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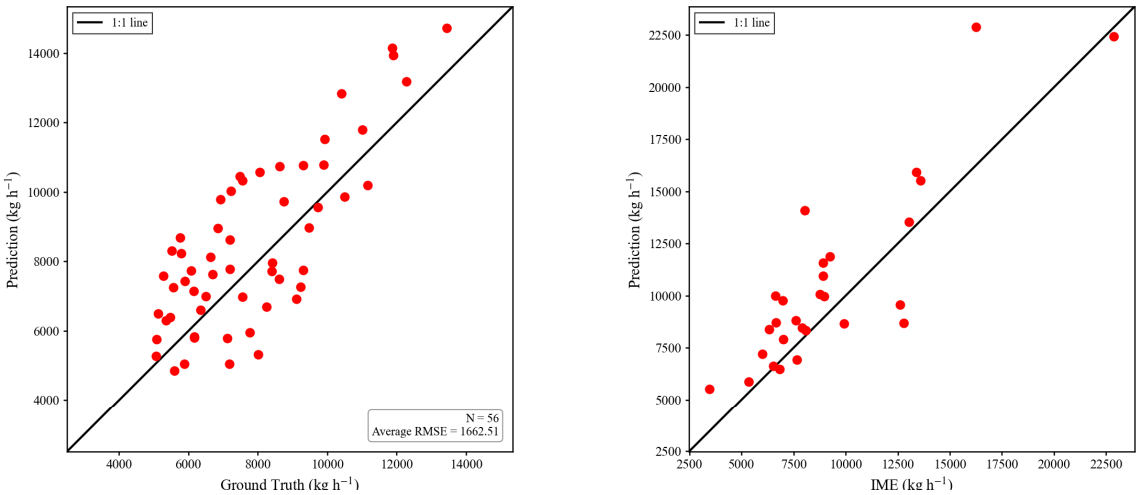


Fig. 9. Scatter plots of methane emission rates predicted by MTU-Former. (a) For the 56 plumes from the public dataset [54]. (b) For the 28 manually collected plumes, cross-validated against the refined IME method [18].

model was trained on the complete dataset. We adopted a stepwise strategy to evaluate the performance bounds of the IME method. First, we conducted tests on noise-free WRF-generated concentration maps to benchmark the theoretical upper limit of the IME method (RMSE: 2349.78 kg h^{-1}). After that, we evaluated the standard threshold-based IME method [18] on the noisy global synthetic dataset. However, the results were suboptimal. Consequently, we refined the IME method by utilizing ground-truth masks from the WRF-LES synthetic plumes and introducing polynomial surface fitting to suppress local background noise. The results are shown in Table IV. We also evaluated performance across different spatial scales. The IME method proved effective in localized regions such as Algeria with an RMSE of 2897.76 kg h^{-1} but degraded to 4323.78 kg h^{-1} on a global scale. These experiments indicate that the IME method is sensitive to plume masks and background noise. Furthermore, as the number of regions increases, calibrating hyperparameters for effective wind speed becomes more difficult, posing a challenge in identifying universal hyperparameters for global application. In contrast to traditional physical integration methods dependent on wind speed, data-driven deep learning models demonstrate greater potential in capturing complex nonlinear mappings from large-scale datasets.

D. Results on the Real-World Dataset

To further assess the practical applicability of MTU-Former in real-world scenarios, we evaluated it on the public dataset released by Mateo-Garcia et al. [54]. From this database, we selected plumes with reported emission rates between 5000 and 30 000 kg h^{-1} and restricted the source imagery to Sentinel-2. Samples with large-area cloud occlusion or incomplete plumes after cropping were removed. This yielded 56 real plumes distributed worldwide, some from regions not covered by our training data. Compared with the emission rates reported in [54], our model achieved an average RMSE of approximately 1662.51 kg h^{-1}, and remained accurate even under slight cloud contamination. In addition, we manually collected 28 plumes from the Permian Basin, Iran, Korpeje, and Hassi Messaoud, and cross-validated our predictions against the IME method [18]. The scatter plots for both parts are shown in Fig. 9, and the visualizations of all plumes are provided in the Supplementary Material.

E. Ablation Experimental Results and Analysis

To assess the efficacy of the core components within the MTU-Former framework while ensuring computational feasibility, we conducted ablation studies using the training and validation sets from the Turkmenistan region as a representative subset. For a fair and reproducible comparison, all ablated variants used the same data split, preprocessing pipeline, training hyperparameters, loss functions, and evaluation metrics as the comparative deep-learning experiments, with only the evaluated component changed. The study focuses on five aspects: comparing the CSWin Transformer backbone with the standard Swin Transformer, examining the base MG-CTFF module in introducing physical consistency, evaluating the methane contribution map and the dual physics-aware attention bias as refinements to the cross-task fusion mechanism, and validating the role of FAMO in balancing heterogeneous task losses. Table V reports the quantitative results.

We first evaluated the impact of the feature extractor. When benchmarking the baseline Swin-UNet against CSWin-UNet without additional modules, the latter reduced the emission rate quantification RMSE from 3948.36 to 3571.97 kg h^{-1}, while increasing the R^2 from 0.8480 to 0.8756. This performance suggests that methane plumes typically exhibit elongated structures. The fixed square windows of the Swin Transformer may limit its ability to model such features. In contrast, the strip-window design of the CSWin Transformer aligns better with these characteristics, enabling efficient aggregation of global context.

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TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF CORE COMPONENTS WITHIN THE MTU-FORMER FRAMEWORK. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Backbone	FAMO Strategy	MG-CTFF Module(base)	Contribution Map	Dual Bias	RMSE (kg h^{-1})	MAPE (%)	R^2	Dice	IoU	Detection Accuracy (%)
Swin-UNet	×	×	×	×	3948.36	23.20	0.8480	0.8235	0.7665	94.31
Swin-UNet	✓	✓	×	×	3456.37	21.15	0.8836	0.8180	0.7611	94.22
CSWin-UNet	×	×	×	×	3571.97	21.66	0.8756	0.8489	0.7936	95.61
CSWin-UNet	✓	×	×	×	3499.06	20.95	0.8807	0.8436	0.7878	**95.67**
CSWin-UNet	×	✓	×	×	3143.64	18.22	0.9037	0.8431	0.7875	95.50
CSWin-UNet	✓	✓	×	×	3090.37	18.43	0.9069	0.8314	0.7749	95.37
CSWin-UNet	✓	✓	✓	×	3027.84	18.00	0.9106	**0.8397**	**0.7839**	95.50
CSWin-UNet	✓	✓	✓	✓	**2997.80**	**17.66**	**0.9124**	0.8323	0.7762	95.37

Next, we verified the effectiveness of the MG-CTFF module in multitask interaction. The integration of this module further reduced the RMSE from 3571.97 to 3143.64 kg h^{-1}, which corresponds to a relative reduction of 12.0%. The MG-CTFF module transforms segmentation outputs into spatial attention, guiding the regression network to prioritize the target region over background noise. Serving as a physical consistency constraint, this mechanism facilitates high-precision point-source quantification.

To ensure more stable training on the global dataset, we examined the impact of the FAMO strategy on model convergence. While the model employing only MG-CTFF attained a satisfactory RMSE of 3143.64 kg h^{-1}, the complete model incorporating the FAMO strategy further reduced the loss metrics. This improvement aligns with the consistent trend observed in other experiments utilizing FAMO. The faster convergence observed with FAMO suggests that gradient competition between tasks may hinder optimization during the later stages of multitask learning. FAMO dynamically adjusts task weights to prevent the faster-converging segmentation task from dominating gradient updates, ensuring continuous optimization of the regression task. This assists the model in overcoming performance limitations and approaching a Pareto optimal state.

Building upon the base MG-CTFF module, we further evaluated two approaches that use physical priors to guide cross-task fusion. Appending the methane contribution map reduced the RMSE from 3090.37 to 3027.84 kg h^{-1} and improved R^2 from 0.9069 to 0.9106, with Dice and IoU also rising to 0.8397 and 0.7839, respectively. This improvement stems from encoding per-pixel methane mass as an explicit input to the mask feature processor, allowing the regression branch to weight each spatial location by its expected mass loading rather than relying solely on binary plume membership. Introducing the dual physics-aware attention bias yielded the lowest RMSE of 2997.80 kg h^{-1} and the highest R^2 of 0.9124, confirming that directly modulating attention scores with plume morphology and methane enhancement priors helps the model concentrate on physically meaningful regions. Dice and IoU decreased slightly to 0.8323 and 0.7762, suggesting that the learnable bias scalars steer the attention distribution toward regression-optimal patterns at the cost of a marginal shift in shared encoder features away from the segmentation objective, though the overall detection accuracy remains nearly unchanged at 95.37%.

The superiority of MTU-Former over the baseline is not only predicated on a single module, as it benefits from the precise capture of plume morphology by the CSWin backbone, the effective utilization of physical boundaries by the MG-CTFF module enhanced with the methane contribution map and dual physics-aware attention bias, and the balancing of training dynamics by the FAMO strategy. Together, these components constitute a lightweight, accurate, and physics-informed multitask network for methane plume analysis.

F. Discussion

MTU-Former showed stable performance on both synthetic data and real Sentinel-2 plume cases, indicating its potential generalizability across different geographic backgrounds and plume morphologies. This generalization is partly supported by the diversity of the training backgrounds. Plume-free Sentinel-2 scenes from 12 regions expose the model to a range of surface reflectance patterns, terrain conditions, and background noise. This helps the model focus on signals related to plumes rather than background features. The background data collected during 2022–2023 also introduces variations in surface reflectance and atmospheric conditions, which provides partial support for seasonal robustness. However, the current assessment of seasonal transferability remains limited by the availability of labeled plume cases. Although the current study focuses on Sentinel-2, the framework is not limited to a single sensor and can be extended to other multispectral or hyperspectral sensors by adapting the retrieval process to their spectral responses, combined with transfer learning or domain adaptation [55].

We further observed that the model's predictions become noticeably less reliable in scenes with complex surface backgrounds, in small plumes with low emission rates, and in plumes with low SNRs. In addition, during the collection of real plume cases, we found that the choice of the MBMP reference image can also introduce substantial prediction errors or even cause missed detections. Representative cases illustrating these effects are shown in Fig. 10.

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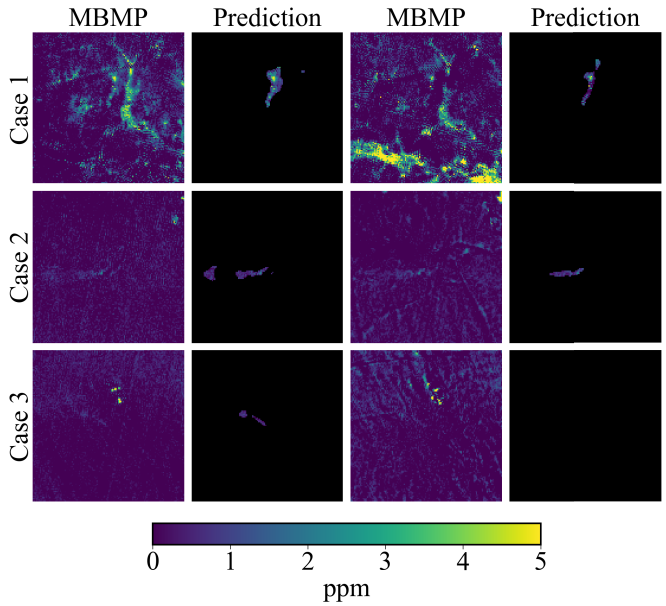


Fig. 10. Effect of the MBMP reference date on plume retrieval. Each row shows the same plume retrieved with two different reference acquisitions. Case 1: Permian Basin (31.7316°N, 102.0420°W), predicted emission rates of 9553.30 (first reference) and 6782.96 kg h^{-1} (second reference). Case 2: Korpeje (38.5575°N, 54.201°E), plume spatial extent varies with reference choice. Case 3: Korpeje (38.4940°N, 54.1977°E), a less suitable reference leads to a missed detection.

For operational deployment, scalability remains an important challenge when applying the framework to million-scale satellite imagery. In large satellite archives, even a low false-positive rate may lead to many plume candidates in scenes without real methane plumes, which would increase the burden of manual inspection and reduce the reliability of routine monitoring. Therefore, large-scale methane monitoring requires not only accurate plume detection, segmentation, and quantification, but also efficient preprocessing, cloud masking, scene filtering, batch inference, and strict false-positive control across large satellite archives. Scalable cloud-based or distributed computation and more efficient data management will also be necessary to support routine monitoring applications. These advancements will serve as a foundation for further optimizing the framework, ultimately progressing toward the establishment of a comprehensive system for global, real-time, and automated methane detection, segmentation, and quantification.

V. Conclusion

In this study, we addressed the challenge of automated global methane monitoring with complex background noise by proposing MTU-Former, a unified multitask deep learning framework that synergizes detection, segmentation, and quantification tasks using large-scale multispectral imagery. By employing a CSWin Transformer backbone to capture complex plume shapes and integrating the MG-CTFF module with the FAMO strategy, the proposed architecture established robust physical consistency and mitigated gradient conflicts, thereby achieving superior noise suppression and an emission rate quantification RMSE of 1532.93 kg h^{-1} without reliance on ancillary meteorological data.

Future work will further enhance MTU-Former in two directions. First, multitemporal observations and attention-based spectral fusion will be incorporated to improve the detection of faint methane plumes. Second, complementary monitoring data sources will be integrated to strengthen the model's capability for global methane monitoring.

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